

CAAM31220 Partial Differential Equations
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Chapter 1

Transport Equation

1.1 Homogeneous transport equation

Theorem 1.1.1 (Homogeneous transport equation). *Consider the initial value problem*

$$\begin{cases} u_t + b \cdot Du = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

If u is a smooth solution, then

$$u(x, t) = g(x - tb).$$

1.2 Inhomogeneous transport equation

Theorem 1.2.1 (Inhomogeneous transport equation). *Consider the initial value problem*

$$\begin{cases} u_t + b \cdot Du = f & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Then

$$u(x, t) = g(x - tb) + \int_0^t f(x + (s - t)b, s) ds.$$

Chapter 2

Laplace Equation

2.1 Mean value property

Definition 2.1.1. A C^2 function satisfying $\Delta u = 0$ is called harmonic.

Theorem 2.1.1 (Mean value property). If $u \in C^2(U)$ is harmonic, then

$$u(x) = \oint_{\partial B(x,r)} u(x) dS = \oint_{B(x,r)} u(x) dx$$

for each $B(x,r) \subset U$.

Theorem 2.1.2 (Converse to the mean value property). If $u \in C^2(U)$ satisfies

$$u(x) = \oint_{\partial B(x,r)} u(x) dS = \oint_{B(x,r)} u(x) dx$$

for $B(x,r) \subset U$, then u is harmonic.

2.2 Maximum principle, positivity, uniqueness, Harnack inequality

Definition 2.2.1. $C(\overline{U}) := \{u \in C(U) : u \text{ is uniformly continuous on every bounded subset of } U\}$. This means u extends continuously to $\overline{U}$.

Theorem 2.2.1 (Maximum principle). Suppose $u \in C^2(U) \cap C(\overline{U})$ is harmonic within U .

1. Then

$$\max_{\overline{U}} u = \max_{\partial U} u.$$

This is the weak maximum principle.

2. Furthermore, let U be connected. If there exists $x_0 \in U$ such that

$$u(x_0) = \max_{\overline{U}} u,$$

then u is constant within U .

This is the strong maximum principle.

Theorem 2.2.2 (Positivity). *Let U be open, bounded, and connected, with $U \subset \mathbb{R}^n$, and let $u \in C^2(U) \cap C(\overline{U})$. Suppose u solves the boundary value problem*

$$\begin{cases} \Delta u = 0 & \text{in } U, \\ u = g & \text{on } \partial U, \end{cases} \quad \text{with } g \geq 0.$$

If $g(z) > 0$ for some $z \in \partial U$, then $u(x) > 0$ for all $x \in U$.

Theorem 2.2.3 (Uniqueness). *Let $g \in C(\partial U)$ and $f \in C(U)$. Then there exists at most one solution $u \in C^2(U) \cap C(\overline{U})$ of*

$$\begin{cases} -\Delta u = f & \text{in } U, \\ u = g & \text{on } \partial U. \end{cases}$$

Definition 2.2.2. $V \subset\subset U$ is defined as $\overline{V}$ is compact and $\overline{V} \subset U$. That is, V is compactly contained in U .

Theorem 2.2.4 (Harnack's inequality). *For each connected open set $V \subset\subset U$, there exists a constant $C = C(V) > 0$ such that*

$$\sup_V u \leq C \inf_V u$$

for all nonnegative harmonic functions u in U .

2.3 Energy method and Dirichlet principle

Theorem 2.3.1 (Dirichlet's principle). *Assume $u \in C^2(\overline{U})$ solves*

$$\begin{cases} -\Delta u = f & \text{in } U, \\ u = g & \text{on } \partial U, \end{cases}$$

the Poisson boundary value problem. Define the energy functional

$$I[w] := \int_U \left(\frac{1}{2} |Dw|^2 - wf \right) dx.$$

Define the admissible set

$$\mathcal{A} := \{w \in C^2(\overline{U}) : w = g \text{ on } \partial U\}.$$

Then

$$I[u] = \min_{w \in \mathcal{A}} I[w].$$

The converse is also true.

2.4 Mathematical supplement: distributions

Definition 2.4.1. For f , $\text{supp}(f) := \overline{\{x \in U : f(x) \neq 0\}}$.

Definition 2.4.2. Define $\mathcal{D}(U) := \{f \in C^\infty(U) : \text{supp}(f) \text{ is compact}\}$. This is the space of test functions on U . $\mathcal{D}(U)$ is equipped with the following notion of convergence: for $(f_n) \subset \mathcal{D}(U)$, $f \in \mathcal{D}(U)$,

$$f_n \rightarrow f \text{ in } \mathcal{D}(U)$$

means that there exists $K \subset\subset U$ such that $\text{supp}(f_n) \subset K$ for all n and $\lim_{n \rightarrow \infty} \|f_n - f\|_{C^k(U)} = 0$ for all k , where

$$\|f\|_{C^k(U)} = \sum_{|\alpha| \leq k} \|D^\alpha f\|_{L^\infty(U)}.$$

Claim 2.4.1.

1. $\mathcal{D}(U)$ is not metrizable.
2. A linear functional

$$T : \mathcal{D}(U) \rightarrow \mathbb{R}$$

is (sequentially) continuous if and only if for every $K \subset\subset U$ there exist $m \in \mathbb{N}$ and $C > 0$ such that, for every $\phi \in \mathcal{D}_K := \{\phi \in \mathcal{D} : \text{supp}(\phi) \subset K\}$,

$$|T(\phi)| \leq C \|\phi\|_{C^m(U)}.$$

3. If $f \in L^1_{\text{loc}} := \{f : U \rightarrow \mathbb{R} : f \text{ measurable, } \int_K |f| dx < \infty \forall K \subset\subset U\}$, then

$$T_f : \mathcal{D}(U) \rightarrow \mathbb{R}, \quad \phi \mapsto T_f(\phi) = \int_U \phi f dx$$

is a sequentially continuous linear functional on $\mathcal{D}$, i.e. $T_f \in \mathcal{D}'$.

4. The map $f \mapsto T_f$ is injective from L^1_{loc} to $\mathcal{D}'$. So $\mathcal{D}'$ contains a copy of L^1_{loc} .

Definition 2.4.3. We call elements in $\mathcal{D}'$ distributions.

Proposition 2.4.2. For all multi-indices α and all $T \in \mathcal{D}'$,

$$D^\alpha T : \phi \mapsto (-1)^{|\alpha|} T(D^\alpha \phi)$$

defines a distribution $D^\alpha T \in \mathcal{D}'$.

Motivation. For $f, \phi \in C_c^\infty(U)$,

$$\int_U (D^\alpha f) \phi dx = (-1)^{|\alpha|} \int_U f (D^\alpha \phi) dx.$$

Proposition 2.4.3. Suppose $u \in C^k(U)$ and α is a multi-index with $|\alpha| \leq k$. If

$$T_u : \phi \mapsto \int_U u \phi dx$$

is the distribution defined by u , then

$$D^\alpha T_u = T_{D^\alpha u} \quad \text{for all } \phi \in \mathcal{D}.$$

That is,

$$(D^\alpha T_u)(\phi) = T_{D^\alpha u}(\phi),$$

or equivalently,

$$(-1)^{|\alpha|} \int_U u D^\alpha \phi = \int_U (D^\alpha u) \phi.$$

2.5 Fundamental solution and representation formulas

Back to Laplace equation

Definition 2.5.1. *The fundamental solution of the Laplace equation is defined by*

$$\Phi(x) = \begin{cases} -\frac{1}{2\pi} \log |x|, & n = 2, \\ \frac{1}{n(n-2)\alpha(n)} \frac{1}{|x|^{n-2}}, & n \geq 3. \end{cases}$$

Proposition 2.5.1. *It is easy to see that $\Phi(x)$ is harmonic away from 0. But it has exactly the right singularity at 0 to produce a point source there:*

$$\int_{\mathbb{R}^n} \Phi(x) \Delta \phi(x) dx = -\phi(0) \quad \forall \phi \in C_c^2(\mathbb{R}^n).$$

That is,

$$\Delta T_\Phi(\phi) = -\delta_0(\phi).$$

Theorem 2.5.2 (Solving Poisson equation). *Define u by*

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y) f(y) dy = \begin{cases} -\frac{1}{2\pi} \int_{\mathbb{R}^2} \log |x-y| f(y) dy, & n = 2, \\ \frac{1}{n(n-2)\alpha(n)} \int_{\mathbb{R}^n} \frac{f(y)}{|x-y|^{n-2}} dy, & n \geq 3. \end{cases}$$

Then

1. $u \in C^2(\mathbb{R}^n)$,
2. $-\Delta u = f$ in $\mathbb{R}^n$.

Definition 2.5.2. *The Green's function for the region U is*

$$G(x, y) = \Phi(y-x) - \phi^x(y),$$

where $\phi^x = \phi^x(y)$ is called a corrector function, which solves the boundary value problem

$$\begin{cases} \Delta \phi^x = 0 & \text{in } U, \\ \phi^x = \Phi(y-x) & \text{on } \partial U. \end{cases}$$

Here $U \subset \mathbb{R}^n$ is open and bounded, and ∂U is of class C^1 .

Theorem 2.5.3 (Representation formula using Green's function). *If $u \in C^2(U)$ solves the Poisson equation*

$$\begin{cases} -\Delta u = f & \text{in } U, \\ u = g & \text{on } \partial U, \end{cases}$$

then

$$u(x) = - \int_{\partial U} g(y) \frac{\partial G}{\partial \nu}(x, y) dS(y) + \int_U f(y) G(x, y) dy.$$

Theorem 2.5.4 (Symmetry of Green's function). *For all $x, y \in U$ with $x \neq y$, we have*

$$G(y, x) = G(x, y).$$

Theorem 2.5.5 (Poisson formula for $\mathbb{R}_+^n$). *Suppose $g \in C(\mathbb{R}^{n-1}) \cap L^\infty(\mathbb{R}^{n-1})$. Let $u : \mathbb{R}_+^n \rightarrow \mathbb{R}$ be defined by*

$$u(x) = \int_{\partial \mathbb{R}_+^n} K(x, y) g(y) dy = \int_{\partial \mathbb{R}_+^n} \frac{2x_n}{n\alpha(n)} \frac{1}{|x - y|^n} g(y) dy.$$

Then

1. $u \in C^\infty(\mathbb{R}_+^n) \cap L^\infty(\mathbb{R}_+^n)$,
2. $\Delta u = 0$ in $\mathbb{R}_+^n$,
3. for every $x_0 \in \partial \mathbb{R}_+^n$, $\lim_{x \rightarrow x_0} u(x) = g(x_0)$.

Theorem 2.5.6 (Poisson formula for a ball). *Suppose $g \in C(\partial B(0, 1))$. Let $u : B(0, 1) \rightarrow \mathbb{R}$ be defined by*

$$u(x) = \int_{\partial B(0, 1)} K(x, y) g(y) dS(y) = \int_{\partial B(0, 1)} \frac{1 - |x|^2}{n\alpha(n)} \frac{1}{|x - y|^n} g(y) dS(y).$$

Then

1. $u \in C^\infty(B(0, 1))$,
2. $\Delta u = 0$ in $B(0, 1)$,
3. for every $x_0 \in \partial B(0, 1)$, $\lim_{x \rightarrow x_0} u(x) = g(x_0)$.

2.6 Mathematical supplement: mollifiers

Definition 2.6.1 (Standard mollifier). *Define $\eta \in C_c^\infty(\mathbb{R}^n)$ by*

$$\eta(x) = \begin{cases} C \exp\left(\frac{1}{|x|^2 - 1}\right), & |x| < 1, \\ 0, & |x| \geq 1, \end{cases}$$

where the constant $C > 0$ is selected so that

$$\int_{\mathbb{R}^n} \eta dx = 1.$$

η is called the standard mollifier.

Definition 2.6.2 (Mollifier). *For $\varepsilon > 0$, set*

$$\eta_\varepsilon(x) = \frac{1}{\varepsilon^n} \eta\left(\frac{x}{\varepsilon}\right).$$

Then the functions η_ε are in $C_c^\infty(\mathbb{R}^n)$ and satisfy

$$\int_{\mathbb{R}^n} \eta_\varepsilon dx = 1, \quad \text{supp}(\eta_\varepsilon) \subset B(0, \varepsilon).$$

Definition 2.6.3 (Mollification). *If $f : U \rightarrow \mathbb{R}$ is locally integrable, define its mollification by*

$$f_\varepsilon = \eta_\varepsilon * f \quad \text{in } U_\varepsilon,$$

that is,

$$f_\varepsilon(x) = \int_U \eta_\varepsilon(x-y)f(y) dy = \int_{B(0,\varepsilon)} \eta_\varepsilon(y)f(x-y) dy.$$

Theorem 2.6.1 (Properties of mollifiers). *Here we have the following properties*

1. $f_\varepsilon \in C^\infty(U_\varepsilon)$.
2. $f_\varepsilon \rightarrow f$ a.e. as $\varepsilon \rightarrow 0$.
3. If $f \in C(U)$, then $f_\varepsilon \rightarrow f$ uniformly on compact subsets of U .
4. If $1 \leq p < \infty$ and $f \in L^p_{\text{loc}}(U)$, then $f_\varepsilon \rightarrow f$ in $L^p_{\text{loc}}(U)$.

2.7 Regularity and further properties

Back to Laplace equation

Theorem 2.7.1 (Smoothness). *If $u \in C(U)$ satisfies the mean value property for each $B(x,r) \subset U$, then $u \in C^\infty(U)$.*

Theorem 2.7.2 (Estimates on derivatives). *Assume u is harmonic in U . Then*

$$|D^\alpha u(x_0)| \leq \frac{C_k}{r^{n+k}} \|u\|_{L^1(B(x_0,r))}$$

for each $B(x_0,r) \subset U$ and each multi-index α with order $|\alpha| = k$, where

$$C_0 = \frac{1}{\alpha(n)}, \quad C_k = \frac{(2^n n)^k}{\alpha(n)}.$$

Theorem 2.7.3 (Liouville's theorem). *Suppose $u : \mathbb{R}^n \rightarrow \mathbb{R}$ is harmonic and bounded. Then u is a constant.*

Theorem 2.7.4 (Representation formula). *For $n \geq 3$, let $f \in C_c^2(\mathbb{R}^n)$. Then any bounded solution of $-\Delta u = f$ in $\mathbb{R}^3$ has the form*

$$u(x) = \int_{\mathbb{R}^n} \Phi(x-y)f(y) dy + C$$

for some constant C .

Remark 2.7.5. The argument fails for $n = 2$, since $\log|x|$ is not bounded at ∞ .

Theorem 2.7.6 (Analyticity). *Assume u is harmonic in U . Then u is analytic in U .*

Chapter 3

Heat Equation

3.1 Fundamental solution and representation formulas

We consider the heat equation

$$u_t - \Delta_x u = 0.$$

Definition 3.1.1 (Fundamental solution of the heat equation). *The fundamental solution of the heat equation on $\mathbb{R}^n$ is the function $\Phi : \mathbb{R}^n \times \mathbb{R} \setminus \{0\} \rightarrow \mathbb{R}$ defined by*

$$\Phi(x, t) = \begin{cases} \frac{1}{(4\pi t)^{n/2}} \exp\left(-\frac{|x|^2}{4t}\right), & x \in \mathbb{R}^n, t > 0, \\ 0, & x \in \mathbb{R}^n, t < 0. \end{cases}$$

Lemma 3.1.1 (Integral of the fundamental solution). *For each $t > 0$,*

$$\int_{\mathbb{R}^n} \Phi(x, t) dx = 1.$$

Theorem 3.1.2 (Solution of the initial-value problem). *Consider the initial value problem*

$$\begin{cases} u_t - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Let $u(x, t)$ be defined by

$$u(x, t) = \int_{\mathbb{R}^n} \Phi(x - y, t) g(y) dy.$$

Then:

1. $u \in C^\infty(\mathbb{R}^n \times \mathbb{R}_+)$,
2. u solves the heat equation,
3. u satisfies the initial value $\lim_{(x,t) \rightarrow (x^0, 0)} u(x, t) = g(x^0)$. for $t > 0$.

Theorem 3.1.3 (Solution of the inhomogeneous problem). *Consider the inhomogeneous problem*

$$\begin{cases} u_t - \Delta u = f, & (t, x) \in \mathbb{R}_+ \times \mathbb{R}^n, \\ u = 0, & (t, x) \in \{0\} \times \mathbb{R}^n. \end{cases}$$

Then

$$u(x, t) = \int_0^t \int_{\mathbb{R}^n} \Phi(x - y, t - s) f(y, s) dy ds.$$

Moreover:

1. $u \in C^\infty$,
2. u solves the PDE,
3. $\lim_{(x,t) \rightarrow (x^0,0)} u(x, t) = 0$.

Theorem 3.1.4 (Solution of the inhomogeneous problem with initial data). *Combining the above two theorems, we discover that*

$$u(x, t) = \int_{\mathbb{R}^n} \Phi(x - y, t) g(y) dy + \int_0^t \int_{\mathbb{R}^n} \Phi(x - y, t - s) f(y, s) dy ds$$

is a solution of

$$\begin{cases} u_t - \Delta u = f & \text{in } \mathbb{R}^n \times (0, \infty), \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}, \end{cases}$$

under the hypotheses on g, f as above (i.e. $g \in C(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$, and $f \in C_c^2(\mathbb{R}^n \times [0, \infty))$ with compact support.)

3.2 Mean value formula

Motivation

Harmonic functions are characterized by averages over balls, where balls arise as level sets of the fundamental solution of $\Delta u = 0$. For the heat equation, we look at level sets of $\Phi(t - s, x - y)$:

$$E(x, t; r) := \left\{ (y, s) \in \mathbb{R}^{n+1} : s \leq t, \Phi(x - y, t - s) \geq \frac{1}{r^n} \right\}.$$

This is called a *heat ball*.

Theorem 3.2.1 (Mean value property for the heat equation). *Let*

$$u \in C_1^2(U_T) := \{u \in C(U_T) : D_x u, D_x^2 u, u_t \in C\}.$$

Then

$$u(x, t) = \frac{1}{4r^n} \iint_{E(x,t;r)} u(y, s) \frac{|x - y|^2}{(t - s)^2} dy ds.$$

Remark 3.2.2. Here C_1^2 means:

- “2”: up to second order derivatives in space,
- “1”: up to first order derivative in time.

3.3 Maximum principle, positivity, uniqueness

Theorem 3.3.1 (Maximum principle). *Here we have*

1. *Weak maximum principle*

$$\max_{\overline{U_T}} u = \max_{\Gamma_T} u.$$

2. *If U is connected and there exists $(x_0, t_0) \in U_T$ such that*

$$u(x_0, t_0) = \max_{\overline{U_T}} u,$$

then u is constant in $\overline{U_{t_0}}$.

This is the strong maximum principle.

Theorem 3.3.2 (Positivity). *Let U be connected, and let $u \in C_1^2(U_T) \cap C(\overline{U_T})$ satisfy*

$$\begin{cases} u_t - \Delta u = 0 & \text{in } U_T, \\ u = 0 & \text{on } \partial U \times [0, T], \\ u = g & \text{on } U \times \{t = 0\}. \end{cases}$$

If $g \geq 0$ and there exists $x_0 \in U$ such that $g(x_0, 0) > 0$, then $u > 0$ everywhere.

Theorem 3.3.3 (Uniqueness on bounded domains). *Let $g \in C(\Gamma_T)$ and $f \in C(U_T)$. Then there exists at most one solution $u \in C_1^2(U_T) \cap C(\overline{U_T})$ of the boundary value problem*

$$\begin{cases} u_t - \Delta u = f & \text{in } U_T, \\ u = g & \text{on } \Gamma_T. \end{cases}$$

Now we study $U = \mathbb{R}^n$

Theorem 3.3.4 (Maximum principle for the Cauchy problem). *Suppose $u \in C_1^2(\mathbb{R}^n \times (0, T)) \cap C(\mathbb{R}^n \times [0, T])$ solves*

$$\begin{cases} u_t - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, T], \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}, \end{cases}$$

and satisfies the growth estimate

$$u(x, t) \leq Ae^{a|x|^2}$$

for constants $A, a > 0$. Then

$$\sup_{\mathbb{R}^n \times [0, T]} u(x, t) = \sup_{\mathbb{R}^n} g(x).$$

Theorem 3.3.5 (Uniqueness for the Cauchy problem). *Let $g \in C(\mathbb{R}^n)$ and $f \in C(\mathbb{R}^n \times [0, T])$. Then there is at most one solution $u \in C_1^2(\mathbb{R}^n \times (0, T)) \cap C(\mathbb{R}^n \times [0, T])$ of the initial value problem*

$$\begin{cases} u_t - \Delta u = f & \text{in } \mathbb{R}^n \times (0, T], \\ u = g & \text{on } \mathbb{R}^n \times \{t = 0\}, \end{cases}$$

satisfying the growth estimate

$$|u(x, t)| \leq Ae^{a|x|^2}$$

for constants $A, a > 0$.

Proof. Apply the theorem on the maximum principle for the Cauchy problem. □

Remark 3.3.6. Without growth control, this is not true.

3.4 Regularity

Definition 3.4.1 (Parabolic Cylinder). *The cylinder is defined as*

$$C(x, t; r) = \{(y, s) \mid |x - y| \leq r, t - r^2 \leq s \leq t\}$$

to denote the closed circular cylinder of radius r , height r^2 , and top center point (x, t) .

Theorem 3.4.1 (Smoothness). *Suppose $u \in C_1^2(U_T)$ solves the heat equation. Then $u \in C^\infty(U_T)$.*

Remark 3.4.2. This is true even if u is not smooth on Γ_T .

Proof. Leverage smoothness of the convolution formula, together with results on cutoffs and uniqueness. $\square$

Theorem 3.4.3 (Estimate on derivatives). *For all $k, \ell \in \mathbb{N}$, there exists $C_{n,k,\ell}$ such that*

$$\max_{Q(x,t;r/2)} \left| D_x^k D_t^\ell u \right| \leq \frac{C_{n,k,\ell}}{r^{k+2\ell+n+2}} \|u\|_{L^1(Q(x,t;r))}$$

for suitable parabolic cylinders $Q(x, t; r)$.

Remark 3.4.4. The norm on $D_x^k D_t^\ell u$ is the Frobenius norm.

3.5 Energy methods and uniqueness

Theorem 3.5.1 (Uniqueness). *There is at most one solution of the boundary value problem*

$$\begin{cases} u_t - \Delta u = f & \text{in } U_T, \\ u = g & \text{on } \Gamma_T, \end{cases}$$

with $u \in C_1^2(\overline{U_T})$.

Theorem 3.5.2 (Backward uniqueness). *Suppose $u, \hat{u}$ solve*

$$\begin{cases} u_t - \Delta u = 0 & \text{in } U_T, \\ u = g & \text{on } \partial U \times [0, T], \end{cases}$$

with $u, \hat{u} \in C_1^2(\overline{U_T})$. Assume at time T we have $u(x, T) = \tilde{u}(x, T)$ for all $x \in U$. Then $u \equiv \hat{u}$ within U_T .

Chapter 4

Wave Equation

4.1 One-dimensional wave equation

We consider $u_{tt} - \Delta u = 0$, and more generally $u_{tt} - \Delta u = f$ (inhomogeneous).

Definition 4.1.1. For $u : \bar{U} \times [0, \infty) \rightarrow \mathbb{R}$, we denote the differential operator

$$\square u := u_{tt} - \Delta u.$$

Theorem 4.1.1 (Solution of the wave equation when $n = 1$). Assume $g \in C^2(\mathbb{R})$, $h \in C^1(\mathbb{R})$. Consider the initial value problem

$$\begin{cases} u_{tt} - u_{xx} = 0 & \text{in } \mathbb{R} \times (0, \infty), \\ u = g & \text{on } \mathbb{R} \times \{t = 0\}, \\ u_t = h & \text{on } \mathbb{R} \times \{t = 0\}. \end{cases}$$

Define u by d'Alembert's formula:

$$u(x, t) = \frac{1}{2} [g(x+t) + g(x-t)] + \frac{1}{2} \int_{x-t}^{x+t} h(y) dy, \quad (x \in \mathbb{R}, t > 0).$$

Then

1. $u \in C^2(\mathbb{R} \times [0, \infty))$,
2. u solves the IVP of the wave equation,
3. $\lim_{(x,t) \rightarrow (x^0, 0)} u(x, t) = g(x^0)$, $\lim_{(x,t) \rightarrow (x^0, 0)} u_t(x, t) = h(x^0)$.

Remark 4.1.2. Factor the differential operator:

$$\square = \partial_t^2 - \partial_x^2 = (\partial_t + \partial_x)(\partial_t - \partial_x).$$

So one can solve a coupled system of two first-order equations:

$$\begin{cases} (\partial_t - \partial_x)u = v, \\ (\partial_t + \partial_x)v = 0. \end{cases}$$

4.2 Higher dimensions: spherical means

Now move to higher dimensions, first consider $n = 3$.

Definition 4.2.1. Let $x \in \mathbb{R}^n$, $t > 0$, $r > 0$. We denote

$$U(x, r, t) := \oint_{\partial B(x, r)} u(y, t) dS(y).$$

Similarly, define

$$G(x, r) := \oint_{\partial B(x, r)} g(y) dS(y),$$

and

$$H(x, r) := \oint_{\partial B(x, r)} h(y) dS(y).$$

Lemma 4.2.1 (Euler–Poisson–Darboux equation). Fix $x \in \mathbb{R}^n$, and let u satisfy the IVP for the n -dimensional wave equation. Then for this fixed x , U is a function of r, t , and we have

$$U \in C^m(\overline{\mathbb{R}_+} \times [0, \infty))$$

and U satisfies

$$\begin{cases} U_{tt} - U_{rr} - \frac{n-1}{r}U_r = 0 & \text{in } \mathbb{R}_+ \times (0, \infty), \\ U = G & \text{on } \mathbb{R}_+ \times \{t = 0\}, \\ U_t = H & \text{on } \mathbb{R}_+ \times \{t = 0\}. \end{cases}$$

Lemma 4.2.2 (Reduction when $n = 3$). For $n = 3$, set

$$\tilde{U} = rU, \quad \tilde{G} = rG, \quad \tilde{H} = rH.$$

Then $\tilde{U}$ solves

$$\begin{cases} \tilde{U}_{tt} - \tilde{U}_{rr} = 0 & \text{in } \mathbb{R}_+ \times (0, \infty), \\ \tilde{U} = \tilde{G} & \text{on } \mathbb{R}_+ \times \{t = 0\}, \\ \tilde{U}_t = \tilde{H} & \text{on } \mathbb{R}_+ \times \{t = 0\}, \\ \tilde{U} = 0 & \text{on } \{r = 0\} \times (0, \infty). \end{cases}$$

Also, $\tilde{U}$ can be expressed by

$$\tilde{U}(x; r, t) = \frac{1}{2} [\tilde{G}(r+t) - \tilde{G}(t-r)] + \frac{1}{2} \int_{-r+t}^{r+t} \tilde{H}(y) dy.$$

Remark 4.2.3. This can be solved by the odd reflection method.

Theorem 4.2.4 (Kirchhoff's formula). For the $n = 3$ IVP for the wave equation,

$$u(x, t) = \frac{\partial}{\partial t} \left(t \oint_{\partial B(x, t)} g dS \right) + t \oint_{\partial B(x, t)} h dS.$$

Equivalently,

$$u(x, t) = \oint_{\partial B(x, t)} (t h(y) + g(y) + Dg(y) \cdot (y - x)) dS(y).$$

Remark 4.2.5.

1. At (x, t) , the solution depends only on g, h evaluated on $\partial B(x, t) \subset \mathbb{R}^3$.
2. The same approach cannot go through on $\mathbb{R}^2$. Instead, we will construct solutions on $\mathbb{R}^2$ by first finding an “ x_3 -free” solution on $\mathbb{R}^3$ and then restricting to $\mathbb{R}^2$.

Theorem 4.2.6 (Poisson formula). *If u is a smooth solution to*

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^2 \times (0, \infty), \\ u|_{t=0} = g & \text{on } \mathbb{R}^2, \\ u_t|_{t=0} = h & \text{on } \mathbb{R}^2, \end{cases}$$

then

$$u(x, t) = \frac{1}{2} \int_{B(x, t)} \frac{t g(y) + t^2 h(y) + t Dg(y) \cdot (y - x)}{(t^2 - |y - x|^2)^{1/2}} dy.$$

Proof. Hadamard’s method of descent. □

Remark 4.2.7. These ideas generalize to higher dimensions; see reading.

Theorem 4.2.8 (Solution of the nonhomogeneous wave equation). *Consider the inhomogeneous problem*

$$\begin{cases} u_{tt} - \Delta u = f, & t > 0, \\ u|_{t=0} = u_t|_{t=0} = 0, & t = 0. \end{cases}$$

Motivated by Duhamel’s principle, define $\tilde{u} = \tilde{u}(x, t; s)$ to be the solution of

$$\begin{cases} \tilde{u}_{tt}(\cdot; s) - \Delta \tilde{u}(\cdot; s) = 0 & \text{in } \mathbb{R}^n \times (s, \infty), \\ \tilde{u}(\cdot; s) = 0 & \text{on } \mathbb{R}^n \times \{t = s\}, \\ \tilde{u}_t(\cdot; s) = f(\cdot, s) & \text{on } \mathbb{R}^n \times \{t = s\}. \end{cases}$$

Set

$$u(x, t) := \int_0^t \tilde{u}(x, t; s) ds.$$

Then

1. $u \in C^2(\mathbb{R}^n \times [0, \infty))$,
2. u is the solution of $u_{tt} - \Delta u = f$ in $\mathbb{R}^n \times (0, \infty)$,
3. $\lim_{(x, t) \rightarrow (x^0, 0)} u(x, t) = 0, \quad \lim_{(x, t) \rightarrow (x^0, 0)} u_t(x, t) = 0.$

4.3 Fourier method

Definition 4.3.1 (Fourier transform and inverse Fourier transform). *For $u \in L^1(\mathbb{R}^n)$, $n \geq 1$, define*

$$\widehat{u}(\xi) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{-ix \cdot \xi} u(x) dx.$$

This is the Fourier transform; denote $\mathcal{F}u = \widehat{u}$. Define also

$$\check{u}(x) = \frac{1}{(2\pi)^{n/2}} \int_{\mathbb{R}^n} e^{ix \cdot \xi} u(\xi) d\xi.$$

This is the inverse Fourier transform; denote $\mathcal{F}^{-1}u = \check{u}$.

Theorem 4.3.1 (Plancherel's theorem). Assume $u \in L^1(\mathbb{R}^n) \cap L^2(\mathbb{R}^n)$. Then $\widehat{u}, \check{u} \in L^2(\mathbb{R}^n)$, and $\|\widehat{u}\|_{L^2(\mathbb{R}^n)} = \|\check{u}\|_{L^2(\mathbb{R}^n)} = \|u\|_{L^2(\mathbb{R}^n)}$.

Remark 4.3.2.

- The Fourier transform preserves the L^2 norm.
- This allows one to define the Fourier transform on L^2 functions by approximating $u \in L^2$ by a sequence $(u_n) \subset L^1 \cap L^2$.
- In fact, the Fourier transform can be defined on the Schwartz space.

Theorem 4.3.3 (Properties of the Fourier transform). Assume $u, v \in L^1(\mathbb{R}^n)$. Then:

1. $\int_{\mathbb{R}^n} u \bar{v} dx = \int_{\mathbb{R}^n} \widehat{u} \bar{\widehat{v}} d\xi$.
2. $(D^\alpha u)^\wedge = (i\xi)^\alpha \widehat{u}$.
3. If $u, v \in L^1 \cap L^2$, then $(u * v)^\wedge = (2\pi)^{n/2} \widehat{u} \widehat{v}$.
4. $u = (\widehat{u})^\vee$.

Theorem 4.3.4 (Solution of the homogeneous wave equation by Fourier transform). Consider the homogeneous IBVP

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u|_{t=0} = g & \text{on } \mathbb{R}^n \times \{t=0\}, \\ u_t|_{t=0} = h & \text{on } \mathbb{R}^n \times \{t=0\}. \end{cases}$$

Then

$$u(t, x) = \cos(t|\nabla|) g + \frac{\sin(t|\nabla|)}{|\nabla|} h,$$

where

$$\cos(t|\nabla|) g := (\cos(t|\xi|) \widehat{g}(\xi))^\vee,$$

and

$$\frac{\sin(t|\nabla|)}{|\nabla|} h := \left(\frac{\sin(t|\xi|)}{|\xi|} \widehat{h}(\xi) \right)^\vee.$$

Proof. Let us do the Fourier transform in the x variable:

$$u(t, x) \mapsto \widehat{u}(t, \xi).$$

Then

$$\begin{cases} \widehat{u}_{tt} + |\xi|^2 \widehat{u} = 0, \\ \widehat{u}|_{t=0} = \widehat{g}, \\ \widehat{u}_t|_{t=0} = \widehat{h}. \end{cases}$$

For each fixed ξ , this leads to an ODE for $\gamma(t) = \widehat{u}(t, \xi)$:

$$\begin{cases} \ddot{\gamma} + |\xi|^2 \gamma = 0, \\ \gamma(0) = \widehat{g}(\xi), \\ \dot{\gamma}(0) = \widehat{h}(\xi). \end{cases}$$

This has solution

$$\gamma(t) = \widehat{g}(\xi) \cos(t|\xi|) + \widehat{h}(\xi) \frac{\sin(t|\xi|)}{|\xi|}.$$

Hence

$$\widehat{u}(t, \xi) = \widehat{g}(\xi) \cos(t|\xi|) + \widehat{h}(\xi) \frac{\sin(t|\xi|)}{|\xi|}.$$

Therefore

$$u(t, x) = (\widehat{u}(t, \xi))^\vee.$$

□

Theorem 4.3.5 (Solution of the inhomogeneous wave equation by Fourier transform). *Consider the inhomogeneous IBVP*

$$\begin{cases} u_{tt} - \Delta u = f & \text{on } \mathbb{R}^n \times (0, \infty), \\ u|_{t=0} = g, \quad u_t|_{t=0} = h & \text{in } \mathbb{R}^n \times \{t = 0\}. \end{cases}$$

Then the problem is solved by

$$u(t, x) = \cos(t|\nabla|) g + \frac{\sin(t|\nabla|)}{|\nabla|} h + \int_0^t \frac{\sin((t-s)|\nabla|)}{|\nabla|} f(s) ds.$$

4.4 Energy method: uniqueness and finite propagation speed

Theorem 4.4.1 (Uniqueness for the wave equation). *Let $U \subset \mathbb{R}^n$ be bounded and open, with smooth boundary ∂U . Consider the following IBVP:*

$$\begin{cases} u_{tt} - \Delta u = f & \text{in } U_T, \\ u = g & \text{on } \Gamma_T, \\ u_t = h & \text{on } U \times \{t = 0\}. \end{cases}$$

Then there exists at most one solution $u \in C^2(\overline{U_T})$.

Theorem 4.4.2 (Finite propagation speed). *Suppose $u \in C^2(\mathbb{R}^n)$ solves*

$$\begin{cases} u_{tt} - \Delta u = 0 & \text{in } \mathbb{R}^n \times (0, \infty), \\ u(x, 0) = u_t(x, 0) = 0 & \text{on } B(x_0, t_0) \times \{t = 0\}. \end{cases}$$

Then $u \equiv 0$ within the cone

$$K(x_0, t_0) = \{(x, t) : 0 \leq t \leq t_0, |x - x_0| \leq t_0 - t\}.$$

Chapter 5

The Method of Characteristics

5.1 Derivation of the characteristic equations

Please review how to use the method of characteristics to solve first-order linear PDEs up to first-order quasi-linear PDEs in MAT351. We now start from the nonlinear equation which is the most general case. Consider the boundary value problem

$$\begin{cases} F(Du, u, x) = 0 & \text{in } U, \\ u = g & \text{on } \Gamma \subset \partial U, \end{cases}$$

where U is open in $\mathbb{R}^n$. The goal is to formulate an equivalent system of ODEs.

Setup

Try to follow paths

$$x(s) = (x^1(s), \dots, x^n(s)), \quad s \in I \subset \mathbb{R},$$

so that we can follow the solution along the curve $s \mapsto x(s)$, starting with initial point $x_0 = x(0) \in \Gamma$. Assume u is a C^2 solution to the PDE. Define

$$z(s) := u(x(s)),$$

and

$$p(s) := Du(x(s)) = \nabla u(x(s)).$$

Definition 5.1.1. *The characteristic equations are defined as*

$$\begin{cases} \dot{p}(s) = -D_x F(p(s), z(s), x(s)) - D_z F(p(s), z(s), x(s))p(s), \\ \dot{z}(s) = D_p F(p(s), z(s), x(s)) \cdot p(s), \\ \dot{x}(s) = D_p F(p(s), z(s), x(s)). \end{cases}$$

Remark 5.1.1. The three equations have the following meanings:

- The equation for $\dot{x}$ chooses the curve $x(s)$ to move in the desired characteristic direction.
- The equation for $\dot{z}$ describes how $u(x(s))$ evolves with respect to the new time variable s .
- The equation for $\dot{p}$ comes from differentiating the PDE with respect to x . Only needed when the PDE is fully nonlinear.

Derivation

Start from the PDE $F(Du, u, x) = 0$. Differentiate the PDE with respect to x^i . We denote the components of F by $F = F(p, z, x)$. Then

$$\sum_{j=1}^n \frac{\partial F}{\partial p^j} \frac{\partial p^j}{\partial x^i} + \frac{\partial F}{\partial z} \frac{\partial z}{\partial x^i} + \frac{\partial F}{\partial x^i} = 0.$$

Equivalently,

$$\sum_{j=1}^n F_{p^j}(Du, u, x) u_{x^i x^j} + F_z(Du, u, x) u_{x^i} + F_{x^i}(Du, u, x) = 0. \quad (*)$$

Now compute

$$\dot{p}^i(s) = \sum_{j=1}^n \frac{\partial p^i}{\partial x^j} \frac{\partial x^j}{\partial s}.$$

we get

$$\dot{p}^i(s) = \sum_{j=1}^n u_{x^i x^j}(x(s)) \dot{x}^j(s). \quad (**)$$

We want to match

$$\sum_{j=1}^n F_{p^j}(Du, u, x) u_{x^i x^j} \text{ with } \sum_{j=1}^n u_{x^i x^j}(x(s)) \dot{x}^j(s).$$

So if we set $\dot{x}^j(s) = F_{p^j}(Du, u, x)$, we have really nice structure. Intuitively, we choose the curve $x(s)$ to move in the direction $D_p F = F_p = (F_{p^1}, \dots, F_{p^n})$. Then equation $(**)$ becomes

$$\dot{p}^i(s) = \sum_{j=1}^n u_{x^i x^j}(x(s)) F_{p^j}(Du, u, x).$$

Plugging this into $(*)$, we get

$$\dot{p}^i(s) + F_z(Du, u, x) u_{x^i} + F_{x^i}(Du, u, x) = 0.$$

Example: First-order linear PDE

Claim 5.1.2. For a first-order homogeneous linear PDE

$$F(Du, u, x) = \langle b(x), Du(x) \rangle + c(x)u(x) = 0,$$

we can solve it by considering the characteristic equations

$$\begin{cases} \dot{x}(s) = b(x(s)), \\ \dot{z}(s) = -c(x(s))z(s). \end{cases}$$

Proof. The function F can be expressed as

$$F(p, z, x) = b(x) \cdot p + c(x)z = 0.$$

Now consider the characteristic equations

$$\begin{cases} \dot{p}(s) = -D_x F(p(s), z(s), x(s)) - D_z F(p(s), z(s), x(s))p(s), \\ \dot{z}(s) = D_p F(p(s), z(s), x(s)) \cdot p(s), \\ \dot{x}(s) = D_p F(p(s), z(s), x(s)). \end{cases}$$

Starting from the equation for $\dot{x}$, since $D_p F = b$, we get $\dot{x}(s) = b(x(s))$. Next, $\dot{z}(s) = b(x(s)) \cdot p(s)$. Using the PDE $b(x) \cdot p + c(x)z = 0$, we obtain $\dot{z}(s) = -c(x(s))z(s)$. Notice that the equation for $\dot{p}$ is not needed here. $\square$

5.2 Mathematical technique: flattening the boundary

Definition 5.2.1. Let $U \subset \mathbb{R}^n$ be open and bounded. We say the boundary ∂U is C^k if, for every $x_0 \in \partial U$, there exist $r > 0$ and a C^k change of coordinates

$$\Phi : U \cap B(x_0, r) \rightarrow \{x \in B(x_0, r) : x^n > \sigma(x^1, \dots, x^{n-1})\}.$$

Likewise, we say ∂U is C^∞ if this holds for $k = 1, 2, \dots$. We say ∂U is analytic if the mapping σ is analytic.

Definition 5.2.2 (Flattening the boundary). The change of coordinates near a point of ∂U is used to flatten out the boundary. Fix $x_0 \in \partial U$, and choose r, σ as above. Define $y = \Phi(x)$ by

$$\begin{cases} y^1 = x^1, \\ \vdots \\ y^{n-1} = x^{n-1}, \\ y^n = x^n - \sigma(x^1, \dots, x^{n-1}). \end{cases}$$

Similarly, set the inverse $x = \Psi(y)$ by

$$\begin{cases} x^1 = y^1, \\ \vdots \\ x^{n-1} = y^{n-1}, \\ x^n = y^n + \sigma(y^1, \dots, y^{n-1}). \end{cases}$$

Then

$$\Phi = \Psi^{-1}, \quad \det D\Phi = \det D\Psi = 1.$$

Claim 5.2.1. After flattening the boundary, rewriting the original PDE for $u(x)$ as a new PDE for $v(y)$ has the form

$$\begin{cases} G(Dv, v, y) = 0 & \text{in } V, \\ v = h & \text{on } \Delta. \end{cases}$$

Proof. Suppose u is a C^1 solution of the boundary value problem

$$\begin{cases} F(Du, u, x) = 0 & \text{in } U, \\ u = g & \text{on } \Gamma \subset \partial U. \end{cases}$$

Rewrite $V := \Phi(U)$, and set $v(y) := u(\Psi(y))$. Equivalently, $u(x) = v(\Phi(x))$. By the chain rule,

$$u_{x^i}(x) = \sum_{k=1}^n v_{y^k}(\Phi(x)) \Phi_{x^i}^k(x).$$

In matrix form,

$$D_x u(x) = D_y v(\Phi(x)) D_x \Phi(x) = \nabla v(\Phi(x))^T D\Phi(x).$$

Thus the PDE written in y variables is

$$F(Dv(y)D\Phi(\Psi(y)), v(y), \Psi(y)) = 0.$$

This is an expression of the form

$$G(Dv(y), v(y), y) = 0 \quad \text{in } V.$$

In addition, $v = h$ on Δ , where $\Delta = \Phi(\Gamma)$ and $h(y) := g(\Psi(y))$. □

5.3 Initial data for the characteristic ODE

If we want to start a characteristic from the boundary point, the starting values cannot be chosen arbitrarily. They must match both the boundary condition and the PDE. After straightening the boundary, we may assume that near $x_0 \in \Gamma$, we have $\Gamma \subset \{x^n = 0\}$. So locally the boundary is just the flat plane $x^n = 0$. Then the boundary condition

$$u = g \quad \text{on } \Gamma$$

becomes locally

$$u(x^1, \dots, x^{n-1}, 0) = g(x^1, \dots, x^{n-1}).$$

In the method of characteristics, one studies curves with unknowns

$$x(s) \in \mathbb{R}^n, \quad z(s) \in \mathbb{R}, \quad p(s) \in \mathbb{R}^n.$$

At time $s = 0$, the characteristic starts with

$$x(0) = x_0, \quad z(0) = z_0, \quad p(0) = p_0.$$

These are the initial conditions for the characteristic ODE, but not every triple (p_0, z_0, x_0) is acceptable.

Definition 5.3.1 (Compatibility conditions). *The following conditions are called the compatibility conditions:*

$$\begin{cases} z_0 = g(x_0), \\ p_0^i = g_{x^i}(x_0), & i = 1, \dots, n-1, \\ F(p_0, z_0, x_0) = 0. \end{cases}$$

A triple

$$(p_0, z_0, x_0) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n$$

verifying these conditions is called admissible.

Remark 5.3.1.

- The value z_0 is uniquely determined by the boundary condition and our choice of boundary point x_0 .
- However, a vector p_0 satisfying the second and third compatibility conditions may not exist, or may not be unique.

5.4 Noncharacteristic boundary condition

If we have an admissible triple (p_0, z_0, x_0) , can we perturb it? For now, assume as above that $x_0 \in \Gamma$, and that Γ near x_0 lies in the flat plane $\{x^n = 0\}$. Set $y = (y^1, \dots, y^{n-1}, 0) \in \Gamma$, with y close to x_0 .

We want to solve the characteristic ODE with initial conditions

$$p(0) = q(y), \quad z(0) = g(y), \quad x(0) = y.$$

Our task now is to find a function $q(\cdot)$ such that $q(x_0) = p_0$ and $(q(y), g(y), y)$ is admissible.

Lemma 5.4.1 (Noncharacteristic boundary conditions). *There exists a unique solution $q(\cdot)$ to the system*

$$\begin{cases} q(x_0) = p_0, \\ q^i(y) = g_{x^i}(y), & i = 1, \dots, n-1, \\ F(q(y), g(y), y) = 0, \end{cases}$$

for all $y \in \Gamma$ close to x_0 , provided that

$$F_{p^n}(p_0, z_0, x_0) \neq 0.$$

Remark 5.4.2.

- The condition $F_{p^n}(p_0, z_0, x_0) \neq 0$ means that the characteristic curve starting from x_0 is not tangent to the flat boundary.

Definition 5.4.1 (Noncharacteristic admissible triple). *In general, if Γ is not flat near x_0 , we say the admissible triple (p_0, z_0, x_0) is noncharacteristic if*

$$D_p F(p_0, z_0, x_0) \cdot \nu(x_0) \neq 0,$$

where $\nu(x_0)$ denotes the outward unit normal vector to Γ at x_0 .

Remark 5.4.3.

- In the MAT351 notes, the PDE is more special:

$$a(x, y, u)u_x + b(x, y, u)u_y = c(x, y, u).$$

Equivalently,

$$F(Du, u, x, y) = a(x, y, u)p_1 + b(x, y, u)p_2 - c(x, y, u) = 0,$$

where $p_1 = u_x$, $p_2 = u_y$. Then $F_{p_1} = a$, $F_{p_2} = b$. So the vector

$$D_p F = (F_{p_1}, F_{p_2}) = (a, b)$$

is the characteristic direction in the (x, y) -plane.

For this quasilinear PDE, the characteristic ODE is

$$\dot{x} = a(x, y, u), \quad \dot{y} = b(x, y, u).$$

So characteristics move in the direction (a, b) . Now the boundary curve is parametrized by $\eta(t) = (k(t), l(t))$. Its tangent vector is $\eta'(t) = (\dot{k}(t), \dot{l}(t))$.

The transversality condition says

$$\det \begin{pmatrix} a(k(t), l(t), g(k(t), l(t))) & \dot{k}(t) \\ b(k(t), l(t), g(k(t), l(t))) & \dot{l}(t) \end{pmatrix} \neq 0.$$

This means the two vectors (a, b) and $(\dot{k}, \dot{l})$ are not parallel. So geometrically, the characteristic direction is not tangent to the boundary. That is exactly what “noncharacteristic boundary” means.

Chapter 6

Mathematical Preliminaries: Function Spaces

6.1 Hölder Spaces

Definition 6.1.1. Let $U \subset \mathbb{R}^n$ be open, let $0 < \gamma \leq 1$, and let $u : U \rightarrow \mathbb{R}$. Suppose u is bounded and continuous. We define

$$\|u\|_{C(\overline{U})} := \sup_{x \in \overline{U}} |u(x)|.$$

Definition 6.1.2 (Hölder continuity). We say that u is Hölder continuous with exponent γ if there exists a constant $C > 0$ such that

$$|u(x) - u(y)| \leq C|x - y|^\gamma$$

for all $x, y \in U$.

Remark 6.1.1.

- Lipschitz continuity is exactly Hölder continuity with exponent is 1.

Definition 6.1.3 (Hölder seminorm and Hölder norm). The γ -th Hölder seminorm of u is defined by

$$[u]_{C^{0,\gamma}(\overline{U})} := \sup_{\substack{x, y \in U \\ x \neq y}} \left\{ \frac{|u(x) - u(y)|}{|x - y|^\gamma} \right\}.$$

The γ -th Hölder norm is defined by

$$\|u\|_{C^{0,\gamma}(\overline{U})} := \|u\|_{C(\overline{U})} + [u]_{C^{0,\gamma}(\overline{U})}.$$

Definition 6.1.4 (Hölder spaces). Let $k \in \mathbb{N}$ and $0 < \gamma \leq 1$. The Hölder space is defined as

$$C^{k,\gamma}(\overline{U}) := \left\{ u \in C^k(\overline{U}) : \|u\|_{C^{k,\gamma}(\overline{U})} < \infty \right\}.$$

The norm is

$$\|u\|_{C^{k,\gamma}(\overline{U})} := \sum_{|\alpha| \leq k} \|D^\alpha u\|_{C(\overline{U})} + \sum_{|\alpha|=k} [D^\alpha u]_{C^{0,\gamma}(\overline{U})}.$$

Theorem 6.1.2. The Hölder space $C^{k,\gamma}(\overline{U})$ is a Banach space.

6.2 L^p Spaces

Definition 6.2.1. Let $U \subset \mathbb{R}^n$ be measurable and let $1 \leq p \leq \infty$. We define

$$L^p(U) := \{f : U \rightarrow \mathbb{R} : f \text{ is measurable and } \|f\|_{L^p(U)} < \infty\}.$$

For $1 \leq p < \infty$, the L^p norm is

$$\|f\|_{L^p(U)} := \left(\int_U |f(x)|^p dx \right)^{1/p}.$$

For $p = \infty$, the L^∞ norm is

$$\|f\|_{L^\infty(U)} := \operatorname{ess\,sup}_{x \in U} |f(x)|.$$

Theorem 6.2.1 (Minkowski inequality). For $1 \leq p \leq \infty$,

$$\|f + g\|_{L^p(U)} \leq \|f\|_{L^p(U)} + \|g\|_{L^p(U)}.$$

Theorem 6.2.2 (Hölder inequality). Let

$$1 \leq p, q \leq \infty, \quad \frac{1}{p} + \frac{1}{q} = 1.$$

Then

$$\|fg\|_{L^1(U)} \leq \|f\|_{L^p(U)} \|g\|_{L^q(U)}.$$

Theorem 6.2.3. The spaces $L^p(U)$ are Banach spaces.

6.3 Weak Derivatives and Sobolev Spaces

Definition 6.3.1 (Weak derivative). Let $u, v \in L^1_{\text{loc}}(U)$, and let α be a multi-index. We say that v is the α -th weak partial derivative of u if

$$\int_U u D^\alpha \phi dx = (-1)^{|\alpha|} \int_U v \phi dx$$

for all test functions $\phi \in C_c^\infty(U)$. We denote this by

$$D^\alpha u = v.$$

Example 6.3.1. See Example 1 and Example 2.

Remark 6.3.2. What is the relation between weak derivative and the distributional derivative?

- Let $\Omega \subset \mathbb{R}^n$ be open and let $u \in L^1_{\text{loc}}(\Omega)$.

A locally integrable function u defines a distribution T_u by

$$T_u(\varphi) = \int_\Omega u(x) \varphi(x) dx, \quad \varphi \in C_c^\infty(\Omega).$$

The *distributional derivative* of u in the x_i -direction is the distribution $\partial_i T_u$ defined by

$$\langle \partial_i T_u, \varphi \rangle = -\langle T_u, \partial_i \varphi \rangle = -\int_\Omega u(x) \partial_i \varphi(x) dx.$$

Thus the distributional derivative is a functional on test functions.

A function $v \in L^1_{\text{loc}}(\Omega)$ is called the *weak derivative* of u in the x_i -direction if

$$\int_{\Omega} v(x) \varphi(x) dx = - \int_{\Omega} u(x) \partial_i \varphi(x) dx$$

for every $\varphi \in C_c^\infty(\Omega)$. In terms of distributions, this condition is

$$T_v = \partial_i T_u.$$

Therefore,

- v is the weak derivative of $u \iff T_v$ is the distributional derivative of T_u .
- weak derivative = function representative of the distributional derivative.
- Not every distributional derivative is represented by a function. For example, let

$$H(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0. \end{cases}$$

Then

$$H' = \delta_0$$

in the sense of distributions, where

$$\delta_0(\varphi) = \varphi(0).$$

Since δ_0 is not given by integration against a locally integrable function, H has a distributional derivative but no weak derivative as an ordinary function.

In summary, Every weak derivative gives a distributional derivative, but not every distributional derivative is a weak derivative.

Proposition 6.3.3 (Uniqueness of weak derivatives). *A weak derivative, if it exists, is uniquely defined up to a set of measure zero.*

Definition 6.3.2 (Sobolev spaces). *Let $U \subset \mathbb{R}^n$ be open, let $1 \leq p < \infty$, $k \in \mathbb{N} \cup \{0\}$. The Sobolev space $W^{k,p}(U)$ consists of all locally summable functions $u : U \rightarrow \mathbb{R}$ such that for each multi-index α with $|\alpha| \leq k$, the weak derivative $D^\alpha u$ exists in the weak sense and belongs to $L^p(U)$.*

Remark 6.3.4. Equivalently,

$$W^{k,p}(U) = \{u \in L^p(U) : D^\alpha u \in L^p(U) \text{ for all } |\alpha| \leq k\},$$

where $D^\alpha u$ is the distributional derivative of u .

Theorem 6.3.5 (Properties of weak derivatives). *Assume $u, v \in W^{k,p}(U)$, with $|\alpha| \leq k$. Then:*

1. $D^\alpha u \in W^{k-|\alpha|,p}(U)$. Moreover,

$$D^\beta(D^\alpha u) = D^\alpha(D^\beta u) = D^{\alpha+\beta} u$$

for all multi-indices α, β with $|\alpha| + |\beta| \leq k$.

2. For all $\lambda, \mu \in \mathbb{R}$,

$$\lambda u + \mu v \in W^{k,p}(U),$$

and

$$D^\alpha(\lambda u + \mu v) = \lambda D^\alpha u + \mu D^\alpha v.$$

3. If V is an open subset of U , then $u \in W^{k,p}(V)$.

4. If $\zeta \in C_c^\infty(U)$, then $\zeta u \in W^{k,p}(U)$, and

$$D^\alpha(\zeta u) = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} D^\beta \zeta D^{\alpha-\beta} u.$$

This is the Leibniz formula.

Remark 6.3.6. Here $\beta \leq \alpha$ means that $\beta_i \leq \alpha_i$ for each $i = 1, \dots, n$, and

$$\binom{\alpha}{\beta} := \frac{\alpha!}{\beta!(\alpha - \beta)!},$$

where $\alpha! = (\alpha_1!) \cdots (\alpha_n!)$.

Proof. Part (4) follows by induction on $|\alpha|$. □

Definition 6.3.3 (Sobolev norm). For $1 \leq p \leq \infty$, define the Sobolev norm by

$$\|u\|_{W^{k,p}(U)} = \begin{cases} \left(\sum_{|\alpha| \leq k} \int_U |D^\alpha u|^p dx \right)^{1/p}, & 1 \leq p < \infty, \\ \sum_{|\alpha| \leq k} \operatorname{ess\,sup}_U |D^\alpha u|, & p = \infty. \end{cases}$$

Example 6.3.7. For $k = 1$ and $1 \leq p < \infty$,

$$\|u\|_{W^{1,p}(U)} = \left(\int_U |u|^p dx + \sum_{i=1}^n \int_U |\partial_i u|^p dx \right)^{1/p} = \left(\|u\|_{L^p(U)}^p + \|Du\|_{L^p(U)}^p \right)^{1/p}.$$

Remark 6.3.8.

- $W^{k,p}(U)$, with $k \geq 1$ and $1 \leq p \leq \infty$, is a Banach space.
- $H^k(U) := W^{k,2}(U)$ is actually a Hilbert space.

6.4 Approximation of Sobolev Functions

Definition 6.4.1 (Convergence in Sobolev spaces). Let $\{u_m\}_{m=1}^\infty \subset W^{k,p}(U)$, $u \in W^{k,p}(U)$. We say that u_m converges to u in $W^{k,p}(U)$ if

$$\lim_{m \rightarrow \infty} \|u_m - u\|_{W^{k,p}(U)} = 0.$$

We denote this by $u_m \rightarrow u$ in $W^{k,p}(U)$.

Definition 6.4.2 (Local Sobolev space). *The local Sobolev space is defined by*

$$W_{\text{loc}}^{k,p}(U) := \left\{ u : u \in W^{k,p}(V) \text{ for every } V \subset\subset U \right\}.$$

We write $u_m \rightarrow u$ in $W_{\text{loc}}^{k,p}(U)$ if $u_m \rightarrow u$ in $W^{k,p}(V)$ for each $V \subset\subset U$.

Theorem 6.4.1 (Local approximation by smooth functions). *Assume $u \in W^{k,p}(U)$ for some $1 \leq p < \infty$. Set $u^\varepsilon := \eta_\varepsilon * u$ in U_ε . Then:*

1. $u^\varepsilon \in C^\infty(U_\varepsilon) \quad \forall \varepsilon > 0$.
2. $u^\varepsilon \rightarrow u$ in $W_{\text{loc}}^{k,p}(U)$ as $\varepsilon \rightarrow 0$.

Theorem 6.4.2 (Meyers–Serrin theorem). *Let $U \subset \mathbb{R}^n$ be open and bounded. Then $C^\infty(U) \cap W^{k,p}(U)$ is dense in $W^{k,p}(U)$.*

Remark 6.4.3.

- This is the global approximation by smooth functions.
- Equivalently, we can say that there exist functions $u_m \in C^\infty(U) \cap W^{k,p}(U)$ such that $u_m \rightarrow u$ in $W^{k,p}(U)$.

Theorem 6.4.4 (Global approximation by functions smooth up to the boundary). *Assume U is bounded and ∂U is C^1 . Suppose $u \in W^{k,p}(U)$ for some $1 \leq p < \infty$. Then there exist functions $u_m \in C^\infty(\bar{U})$ such that $u_m \rightarrow u$ in $W^{k,p}(U)$.*

Theorem 6.4.5 (Extension theorem). *Assume U is bounded and ∂U is C^1 . Select a bounded open set V such that $U \subset\subset V$. Then there exists a bounded linear operator*

$$E : W^{1,p}(U) \rightarrow W^{1,p}(\mathbb{R}^n)$$

such that for each $u \in W^{1,p}(U)$, we have:

1. $Eu = u$ a.e. in U ;
2. $\text{supp}(Eu) \subset V$;
3. $\|Eu\|_{W^{1,p}(\mathbb{R}^n)} \leq C\|u\|_{W^{1,p}(U)}$, where the constant C depends only on p , U , and V .

Definition 6.4.3. *We call Eu an extension of u to $\mathbb{R}^n$.*

6.5 Embeddings Between Sobolev Spaces

If $f \in W^{k,p}(U)$, does this automatically imply that f belongs to any other spaces?

Theorem 6.5.1 (Gagliardo–Nirenberg–Sobolev inequality). *Assume $1 \leq p < n$. Then there exists a constant $C = C(n, p)$ such that, for all $u \in C_c^1(\mathbb{R}^n)$, we have*

$$\|u\|_{L^q(\mathbb{R}^n)} \leq C\|Du\|_{L^p(\mathbb{R}^n)},$$

where $\frac{1}{q} + \frac{1}{n} = \frac{1}{p}$. Equivalently, $q = \frac{np}{n-p}$.

Corollary 6.5.2 (Estimate for $W^{1,p}$, $1 \leq p < \infty$). *Let $U \subset \mathbb{R}^n$ be bounded and open, and suppose ∂U is C^1 . Assume $1 \leq p < n$ and $u \in W^{1,p}(U)$. Then $u \in L^q(U)$, with the estimate*

$$\|u\|_{L^q(U)} \leq C \|u\|_{W^{1,p}(U)}.$$

Definition 6.5.1. *Let $U \subset \mathbb{R}^n$ be open and bounded. We denote*

$$W_0^{1,p}(U) := \overline{C_c^\infty(U)}^{W^{1,p}(U)}.$$

That is, $W_0^{1,p}(U)$ is the closure of $C_c^\infty(U)$ in the $W^{1,p}(U)$ norm.

Corollary 6.5.3 (Estimate for $W_0^{1,p}$, $1 \leq p < \infty$). *Let $U \subset \mathbb{R}^n$ be bounded and open. Suppose $u \in W_0^{1,p}(U)$ for some $1 \leq p < n$. Then we have the estimate*

$$\|u\|_{L^q(U)} \leq C \|Du\|_{L^p(U)}$$

for each $q \in \left[1, \frac{np}{n-p}\right]$, where the constant $C = C(q, p, n, U)$.

Remark 6.5.4.

- For $n = 1$, the restriction $p < n$ is necessary. For example,

$$x \mapsto \log \left(\log \left(1 + \frac{1}{|x|} \right) \right)$$

belongs to $W^{1,n}(B_1)$, but not to $L^\infty(B_1)$.

- For the case $n > p$, we use Morrey's inequality.